

Vision & Scope Document

Smart Career Platform (CareerHub) - Report 1

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1. Product Background

Today's recruitment platforms, including Applicant Tracking Systems (ATS) and online job portals, have significantly improved the way companies manage recruitment workflows. They help recruiters publish job postings, collect applications, track candidates throughout the hiring process, and perform basic resume screening. These capabilities reduce administrative work and make recruitment more organized.

However, when it comes to evaluating how well a candidate matches a specific job, existing systems still face important limitations. Most platforms primarily rely on automated screening techniques that cannot fully understand the context of a candidate's experience or accurately assess whether the candidate satisfies the key requirements of a position. As a result, applicants with similar resumes may receive very different evaluations, while highly qualified candidates can be overlooked because their experience or skills are expressed differently from the job description.

Although recent recruitment platforms have incorporated AI-powered matching features, many of these systems still produce a single matching score without clearly explaining how the result is obtained. Recruiters therefore cannot easily determine whether the score is driven by technical skills, work experience, education, or other important qualifications. Consequently, they still need to manually review shortlisted CVs to verify the candidate's actual suitability for the role.

The result is that recruitment teams continue to spend a significant amount of time comparing CVs with job descriptions, manually validating candidate qualifications, and making ranking decisions based largely on personal judgment. Existing systems improve administrative efficiency but have not fully addressed the core challenge of providing an objective, transparent, and reliable evaluation of candidate-job fit.

This gap motivates the proposed system. Instead of relying on a single overall matching score, the system evaluates candidate-job fit from multiple perspectives and provides a more interpretable ranking of applicants. By offering a structured and transparent assessment, the system aims to support recruiters in making faster, more consistent, and more reliable hiring decisions while reducing the amount of manual effort required during the screening process.

2. Existing Solutions

2.1 Indeed

2.1.1. What is Indeed?

- Website/Link: [Indeed](#)
- Description: Indeed is an online recruitment platform that allows employers to post jobs, manage applicants, and review candidate profiles through an employer dashboard. Its candidate management tools support basic applicant organization, filtering, sorting, and status management during the hiring process. According to Indeed's employer resources, recruiters can filter applicants by criteria such as location, screener question status, interest level, and application date.

2.1.2. How recruiting in Indeed works

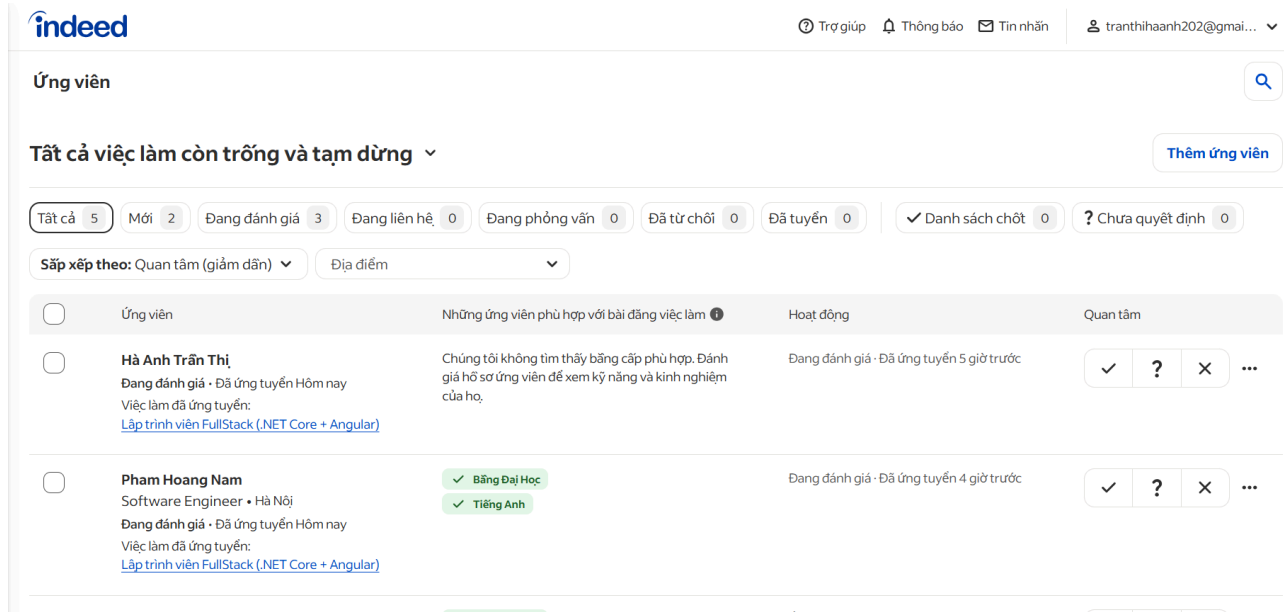


Figure 2.1.2.1. CV screening in Indeed

a. The candidate screening process in Indeed is mainly based on candidate management, basic filtering, sorting, and manual HR review. The workflow generally follows these steps:

- **Job Posting:** Employers create a job post with basic information such as job title, job description, location, and job requirements. During this step, recruiters can also create screening questions related to education, language ability, work experience, certificates, or other required conditions.
- **Candidate Application:** Candidates apply for the job by submitting their profiles or CVs. They are also required to answer the screening questions created by the recruiter as part of the application process.
- **Candidate List Management:** After submission, Indeed displays candidates in the employer dashboard. Recruiters can view and manage applicants based on candidate name, job applied for, location, application date, and status.
- **Filtering and Sorting:** Recruiters can apply basic filters and sorting options to organize the candidate list. The filtering process can be modeled as:
 $Candidate\ Data + Selected\ Filters \rightarrow Filtered\ Candidate\ List \rightarrow Recruiter\ Review$
- **Suitability Tags:** Indeed may display tags such as Degree or English in the “suitable candidates” column. However, these tags appear to be based on candidates’ answers to recruiter-created screening questions, not a deep analysis of the whole CV. This process can be modeled as:
 $Recruiter\ Screening\ Questions \rightarrow Candidate\ Answers \rightarrow Suitability\ Tags \rightarrow Candidate\ List\ Display$

b. Limitations and Practical Observations

Based on practical testing and observation, Indeed has several limitations:

- **Basic Filtering Only:** Indeed mainly supports simple filtering and sorting functions, such as candidate name, job applied for, location, application date, and status. These functions help organize candidates but do not deeply evaluate CV content.
- **Suitability Tags Depend on Screening Questions and Lack Filtering:** The “suitable candidates” column is mainly based on answers to HR-created screening questions, reflecting candidate responses rather than a complete analysis of skills,

experience, projects, or job-specific keywords in the CV. Additionally, although these tags are displayed, there is no separate filter for this column, meaning recruiters cannot directly filter candidates based on these tags.

- **Manual CV Review Still Required:** Because the system does not deeply analyze CV content, recruiters still need to manually open and review each CV to evaluate skills, work experience, education, and job-specific requirements.

Overall, Indeed can be considered a search-and-filter-based recruitment solution rather than a transparent AI-based CV screening system.

2.2 TopDev

2.2.1. What is TopDev

- Website/Link: <https://topdev.vn/>
- Description: TopDev is a leading technology-driven recruitment and HR platform in Vietnam. It primarily operates as an online CV builder and job board, connecting millions of job seekers with hundreds of thousands of employers.

2.2.2. How recruiting in TopDev works

⚡ Show best matched candidates

No	Candidate information	Candidate experience	Job title	Action
1	H Ha Anh Tran.Thi tranthihaanh202@gmail.com 0829260369 Best Match + Has cover letter Applied date: 30-06-2026	Experience: 2022/01 - 2024/12 Gia sư - GIA SƯ TOÁN LỚP 9 Education: 2022/01 - 2026/12 ĐẠI HỌC FPT	Lập trình viên Backend .NET	Select procedure
2	V Võ.Ta.Dong tadong2003@gmail.com 0945767510 Best Match + Applied date: 30-06-2026	Skill(s): Java, C#, MVC, Github, SQL Server, Git, Visual Studio, My SQL, PostgreSQL, Postman Experience: 2023/04 - 2023/08 Inter,NET BackEnd Developer - FPT SOFTWARE 3 YOE 2025/11 - 2026/04 Inter,NET BackEnd Developer - FOX.AI Education: FPT UNIVERSITY	Lập trình viên Backend .NET	Select procedure
3	N Nguyen Duc.Tien carina.ndt@gmail.com 0982872004 Has cover letter Applied date: 30-06-2026	Skill(s): CSS, C++, JavaScript, HTML, MongoDB, Git, ReactJS, ExpressJS Experience: 2026/04 - 2026/06 AI Engineer - VinSmart Pay 2025/08 - 2026/04 Full-Stack Developer - Zamiga Joint Stock Company Education: 2026/04 - Present Thang Long University	Lập trình viên Backend .NET	Select procedure

Total 3 items

Figure 2.2.2.1. CV screening in TopDev

$$JD_k$$

$$C = \{c_i \mid i = 1 \dots N\}$$

$$\text{Matching}(JD_k, C) \rightarrow \text{set } M \subseteq C$$

- The candidate screening process in TopDev is primarily automated, focusing on data extraction and profile matching. The workflow generally follows these steps:
 - **CV Parsing:** When candidates submit their CVs, the system employs automated tools to parse the document and extract key entities, including technical skills, work experience, and education details.
 - **Automated Ranking:** The system uses these extracted entities to compare the candidate's profile against specific job requirements and applies a ranking mechanism to identify potential candidates.
- Limitations and Practical Observations
 - Based on practical testing and observation, the system exhibits several significant shortcomings:
 - **Inconsistent Skill Extraction:** The system's ability to parse technical skills from CVs is often unstable. It frequently fails to recognize key technical competencies or miscategorizes non-technical terms as core skills.

- **Lack of Granular Metrics:** The ranking system relies on a simplistic binary logic, categorizing candidates solely as either "Match" or "No-match." Unlike systems that provide a percentage-based compatibility score (e.g., 85% match), TopDev fails to provide a nuanced ranking, leaving recruiters without a clear way to prioritize candidates based on the strength of their qualifications.
- **Lack of Explainability:** The system does not offer any rationale or transparency behind its matching decisions. Recruiters are not informed why a candidate is tagged as a "match," which complicates the verification process and reduces trust in the automated results.
- **Contextual Inaccuracy:** Because the system appears to lack a deep understanding of professional context, it frequently produces false positives. For example, in observed scenarios, candidates with irrelevant professional backgrounds—such as a math tutor—have been incorrectly flagged as a "best match" for senior technical roles like a .NET developer. This highlights a failure to effectively distinguish between generic skill keywords and specific, relevant industry experience.

2.3 TopCV

2.3.1 What is TopCV

- Website/Link: [TopCV](#)
- TopCV is an online recruitment platform that supports job searching, application management, and AI-based candidate evaluation. The platform provides CV Scoring Beta for candidate classification and CV Scoring Pro for scoring and criterion-based evaluation. However, based on literature review and practical observation, TopCV currently supports AI-based CV evaluation only for recruiters and does not provide CV-to-Job Description comparison or personalized CV improvement recommendations for candidates.

2.3.2. How recruiting in TopCV works



Figure 2.3.2.1. CV Scoring Beta in TopCV

- The candidate evaluation process in TopCV is mainly based on AI-assisted CV analysis and candidate suitability assessment. The workflow of CV Scoring Beta generally follows these steps:
 - **Job Posting:** Recruiters create a job posting containing information such as the job title, job description, recruitment requirements, and professional domain.
 - **Candidate Application:** Candidates apply for the job by submitting their CVs through the TopCV platform.
 - **AI-based CV Analysis:** After receiving applications, the system analyzes the candidate's CV together with the job description to evaluate the candidate's suitability for the position.
 - **CV Scoring Beta:** In the CV Scoring Beta mode, the system classifies candidates into one of three predefined categories: Suitable, Consider, or Not Suitable. This process can be modeled as:

CV + Job Description → AI Evaluation → Suitable / Consider / Not Suitable

- **Result Display:** The evaluation results are displayed in the recruiter dashboard to support candidate screening and selection. However, the system does not provide detailed scores or explanations for the assigned suitability category, which limits transparency and makes it difficult for recruiters to distinguish candidates within the same category.

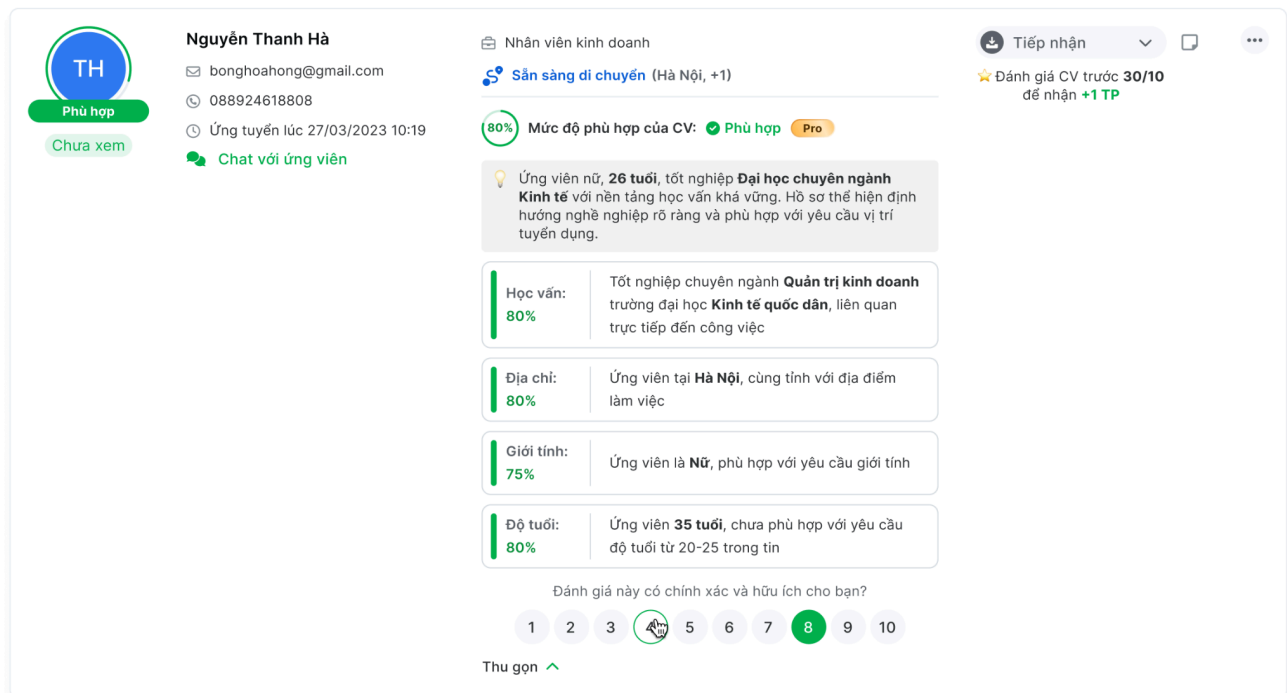


Figure 2.3.2.2. CV Scoring Pro in TopCV

The workflow of CV Scoring Pro generally follows these steps:

- **CV Scoring Pro:** In the CV Scoring Pro mode, the system provides additional evaluation information, including an overall matching score, AI-generated comments, criterion-based evaluation scores, and explanations for each evaluation criterion. Based on practical observation, the evaluation criteria include educational background, age, gender, location, and job position suitability. This process can be modeled as:

Candidate CV + Job Description → AI Evaluation → Overall Score + Criterion-based Scores + AI-generated Explanations → Recruiter Dashboard

- **Evaluation Criteria Assessment:** The system evaluates candidates based on predefined criteria and generates corresponding scores and explanations. However, the scoring mechanism and the weighting of individual criteria are not disclosed, reducing the transparency and explainability of the evaluation process.
- **Domain Limitation:** The CV Scoring feature is currently available only for a limited number of predefined professional domains and job categories, which restricts the scalability and generalizability of the system.

From the candidate's perspective, the recruitment process follows a traditional workflow consisting of the following steps:

Job Search → View Job Description → Submit CV → Wait for Recruiter Response

- **Candidate Support Features:** The system does not provide functionalities such as CV-to-Job Description comparison, missing skill identification, personalized CV improvement recommendations, or suitability prediction before application submission.
- b. Limitations and Practical Observations
 - Based on literature review and practical observation, although TopCV provides AI-based CV evaluation features for recruiters, the system still exhibits several significant limitations:
 - **Limited Transparency in CV Scoring Beta:** The CV Scoring Beta version only classifies candidates into three categories: Suitable, Consider, or Not Suitable without providing detailed scores or explanations. This limitation reduces transparency and makes it difficult for recruiters to understand the rationale behind the evaluation results.
 - **Limited Explainability and Potential Bias:** Although CV Scoring Pro provides an overall matching score and criterion-based evaluations, the scoring mechanism and the weighting of individual criteria are not fully disclosed. Furthermore, the use of evaluation criteria such as education, age, gender, and location may introduce bias and does not always accurately reflect a candidate's actual ability to perform the job.
 - **Limited Domain Coverage:** The CV Scoring functionality currently supports only a limited set of predefined professional domains and job categories. This restriction reduces the scalability and generalizability of the AI evaluation system when applied to new industries or emerging job roles.
 - **Lack of Flexible Ranking and Customization:** The system does not support candidate ranking based on customizable evaluation criteria for different job positions. Recruiters are also unable to adjust the weights of individual evaluation factors, reducing both the flexibility and explainability of the screening process.
 - **Lack of Candidate-Centric Analysis:** From the candidate's perspective, the system does not provide functionalities for analyzing the compatibility between a CV and a Job Description, identifying missing skills, or generating personalized CV improvement recommendations. As a result, candidates cannot effectively identify competency gaps or optimize their CVs for specific job opportunities.

2.4 Summary of Existing Solutions

Current recruitment platforms such as Indeed, TopDev, and TopCV adopt different approaches to support the recruitment and candidate screening process. Indeed primarily focuses on candidate search, filtering, and management mechanisms; TopDev applies CV parsing techniques combined with automated profile matching; while TopCV utilizes AI-based CV scoring to support candidate evaluation. Although these systems have contributed to partially automating the recruitment process, they still exhibit several significant limitations.

First, many existing systems mainly support candidate management, search, or classification based on predefined criteria, rather than performing a comprehensive analysis of the compatibility between a candidate's CV and the Job Description (JD). As a result, recruiters still need to spend considerable time manually reviewing important factors such as skills, work experience, projects, and job-specific requirements.

Second, systems that rely on CV parsing are highly dependent on the accuracy of the information extraction process. Errors such as missing skills, incorrect identification of professional expertise, or inaccurate classification of work experience can directly affect candidate

evaluation, matching, and ranking processes. These limitations may lead to both false positive and false negative results, thereby reducing the overall reliability of the system.

Third, existing AI-based CV evaluation systems often employ relatively limited evaluation criteria and do not fully reflect a candidate's actual suitability for a specific job position. Many evaluation criteria are not directly related to job requirements, while factors such as age, gender, or location may introduce bias into the evaluation process. In addition, current systems typically lack multidimensional evaluation mechanisms, customizable weighting strategies, and transparent explainability methods for their assessment results.

Furthermore, most existing platforms do not support candidates in analyzing the gap between their CV and the Job Description, identifying missing skills, or providing recommendations for improving their profiles before applying for a job. Therefore, although current recruitment solutions have incorporated automation and artificial intelligence technologies, significant limitations still exist in terms of accuracy, comprehensiveness, explainability, flexibility, and decision-support capabilities for both recruiters and job seekers.

2.5 Opportunity for Our System

Given the limitations of current recruitment platforms, researching and developing a next-generation recruitment system presents strategic opportunities to reshape the entire candidate evaluation process. Rather than merely serving as a passive data repository and filtering tool, a system utilizing Natural Language Processing (NLP) and Generative AI can function as an intelligent analytical assistant. This new solution not only overcomes the core drawbacks of legacy technologies but also delivers breakthrough value in resource optimization and enhanced recruitment accuracy.

First, the increasing volume of job applications creates an opportunity to automate the initial screening process and reduce recruiters' manual workload. Instead of manually reviewing CVs, comparing candidate qualifications with job descriptions, and ranking applicants, an AI-assisted recruitment system can convert unstructured CVs and JDs into structured representations, evaluate candidates across multiple job-related dimensions, and generate transparent ranking results. This enables HR professionals to spend more time on interviews and hiring decisions while significantly improving screening efficiency.

Second, recent advances in natural language processing and large language models provide an opportunity to improve the accuracy and reliability of candidate evaluation. By combining structured information extraction, semantic embeddings, skills matching, experience analysis, education verification, and requirement checking within a multi-dimensional scoring framework, the proposed system can assess candidate-job compatibility more comprehensively than a single matching approach. In addition, configurable criterion weights and detailed score breakdowns allow recruiters to adapt the evaluation process to different job positions while maintaining transparency and explainability.

Finally, AI-based recruitment systems create opportunities beyond candidate ranking by supporting long-term talent management and fair recruitment practices. The proposed system can transform parsed CVs into a structured candidate knowledge base, allowing recruiters to efficiently retrieve previously evaluated candidates for future positions without relying on manual records or memory. By evaluating only job-related qualifications and excluding sensitive personal information, the system also promotes more objective and consistent hiring decisions while providing scalable AI-assisted decision support for future recruitment activities.

3. Proposed Solution

3.1 Overview

The AI CV Ranking system evaluates the compatibility between a candidate's CV and a specific job description. It does not score a CV based on visual design or general impression, but focuses on how well the CV satisfies job-related requirements. According to the *Summary of the Uniform Guidelines on Employee Selection Procedures*, selection procedures may include interviews, reviews of experience, education, tests, work samples, and other methods used for employment decisions. When a selection procedure is used to rank candidates, there should be evidence that higher scores are related to better job performance [1]. Therefore, the system should evaluate candidates based on job-related criteria such as skills, experience, education, and specific JD requirements, while excluding sensitive or irrelevant information such as gender, age, photo, marital status, religion, race, hometown, or personal background.

CVs are usually unstructured documents, and candidates may describe the same skills or experience in different ways. Risavy et al. state that standardized application forms can support more consistent comparison of job-related information across applicants [2]. Therefore, the system first converts CV and JD content into structured data before scoring. Previous NLP research also supports AI-based CV-JD matching. Li et al. used 6,492 resumes from 24,933 applications for 252 positions and achieved 79.2% performance in resume-job matching using transformer-based models [3]. However, semantic similarity alone may miss required skills, certificates, degrees, or minimum experience, while keyword matching alone may miss synonyms, abbreviations, or related terms. Therefore, the proposed system uses a multi-dimensional scoring model instead of relying on a single matching method.

The system follows three main stages: parsing, embedding, and scoring. In the parsing stage, CV and JD text are converted into structured JSON data. In the embedding stage, the system converts the content into vectors to capture semantic meaning. In the scoring stage, the system calculates five dimension scores and a final score from 0 to 100 [5]. The five evaluation dimensions are summarized below:

Dimension	Criterion	Description
D1	Semantic Similarity	Measures semantic similarity between CV and JD using embeddings and cosine similarity.
D2	Skills Matching	Measures how well candidate skills match required and preferred JD skills.
D3	Experience Relevance	Evaluates years of experience, relevance, recency, and employment gaps.
D4	Education Match	Checks whether education satisfies the job requirement.

D5	Location match	Checks whether or not
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To improve flexibility, HR can customize the weight of each criterion depending on the job position. For example, a software developer role may prioritize technical skills, while a management role may prioritize relevant experience and domain knowledge. The final score is calculated as follows:

$$\text{final} = \left(\sum_{i=1}^5 D_i W_i \right) \times 100$$

Where D1 to D5 are the dimension scores, and W1 to W5 are the weights configured by HR. The system can also apply penalties for critical missing requirements, such as missing must-have skills or insufficient experience. According to the NIST AI Risk Management Framework, trustworthy AI should be transparent, explainable, reliable, privacy-enhanced, and fair [4]. Therefore, the system provides not only a final score but also a detailed score breakdown to support HR decision-making, while the final hiring decision remains under human review.

3.2 Five-Dimensional Scoring Engine

3.2.1. D1: Semantic Similarity (Default weight: 30%)

$$r = \text{cosine}(\vec{c}, \vec{d})$$

$$D_1 = \max \left(0, \min \left(\frac{r - r_{\min}}{r_{\max} - r_{\min}}, 1 \right) \right) = \max \left(0, \min \left(\frac{r - 0.20}{0.80 - 0.20}, 1 \right) \right)$$

- r is raw cosine similarity
- $r_{\min} = 0.20$, $r_{\max} = 0.80$ because real CV-JD pairs rarely go below 0.20 or above 0.80
- Normalized range: $[0.20, 0.80] \rightarrow [0, 1]$
- Model: all-MiniLM-L6-v2 (384-dim) or text-embedding-3-small (1536-dim)

3.2.2. D2: Skill Match (Default weight: 35%)

$$D_2 = \frac{\sum_{i=1}^{|S_{JD}|} w_i \cdot m_i}{\sum_{i=1}^{|S_{JD}|} w_i}$$

- i ranges over each skill in the JD's required_skills list
- w_i importance weight of skill i (1 = nice to have, 2 = preferred, 3=must-have)
- m_i is the match quality of skill i against the CV

Fuzzy Matching (Ratcliff-Obershelp):

$$\text{ratio}(a, b) = \frac{2 \times |\text{matching blocks}|}{|a| + |b|}$$

- a, b = two skill strings being compared (e.g. JD's "postgresql" vs CV's "postgres")
- "matching blocks" = total length of the longest common contiguous substrings found recursively (Gestalt pattern matching, not edit distance)
- Threshold ≥ 0.85 to accept as a fuzzy match

Category Partial Credit:

- C_k = the category (e.g. "FE frameworks") that the JD skill S_{JD} belongs to
- $S_{cv} \cap C_k$ = how many skills from that same category the candidate has
- Base credit is 0.3 for having any related skill; each additional skill in the same category adds +0.1, capped at 0.5
- This rewards "adjacent" expertise (e.g. JD wants Vue, candidate knows React+Angular) without letting it fully substitute for the exact skill

$$\text{cat_score} = \min(0.3 + 0.1 \times (|S_{cv} \cap C_k| - 1), 0.5)$$

3.2.3. D3 - Experience (Default weight - 20%)

Base ratio with contextual modifiers

$$\text{base} = \min\left(\frac{Y_{cv}}{Y_{jd}}, 1.0\right)$$

$$D_3 = \text{clamp}(\text{base} + \delta_r + \delta_t + \delta_o, 0, 1)$$

- Y_{cv} = candidate's total years of experience; Y_{jd} = JD's minimum required years
- δ_r = relevance bonus, up to +0.20 scaled by how many of those years were in a relevant domain
- δ_t = recency modifier, +0.10 if currently employed, -0.10 if there's a > 12-month gap
- δ_o = over-qualification penalty -0.05 if $Y_{cv} > 2 \times Y_{jd}$ (JD rejects over-senior candidates)
- clamp: final result is forced back into [0,1]

3.2.4. D4 - Education (Default weight - 10%)

$$D_4 = \min\left(\frac{L_{cv}}{L_{jd}}, 1.0\right)$$

Degree mapping: High School=1, Associate=2, Bachelor=3, Master=4, PhD=5

3.2.5. D5 - Keywords (Default weight - 5%)

- t is the estimated driving time (in minutes) between the candidate's address and the JD's office address.

Work-mode compatibility multiplier M :

3.2.6. Final score

$$\text{final} = (D_1W_1 + D_2W_2 + D_3W_3 + D_4W_4 + D_5W_5) \times 100$$

Default weights:

- $W_1 = 0.30$ (Semantic)
- $W_2 = 0.35$ (Skills)

- $W_3 = 0.20$ (Experience)
- $W_4 = 0.10$ (Education)
- $W_5 = 0.05$ (Keywords)

Hard rule penalties

$$P_{\text{skill}} = \min(0.20|M|, 0.70)$$

$$P_{\text{exp}} = \begin{cases} 0.30 & Y_{CV} < 0.8Y_{JD} \\ 0 & \text{else} \end{cases}$$

$$P_{\text{total}} = \min(P_{\text{skill}} + P_{\text{exp}}, 0.95)$$

$$\text{final}_{\text{penalized}} = \text{final} \times (1 - P_{\text{total}})$$

- $|M|$ = count of missing must-have skills (weight=3)

4. Project Scope & Limitations

4.1 Major Features

ID	Feature Name	Description	Addressed Gap
FE-01	User & Authentication Management	Provides registration, login, role-based access control, password recovery, and account management for Candidates, HRs, and Admins to ensure secure access to the recruitment platform.	
FE-02	Company & HR Verification Management	Enables company profile verification, HR account approval, verification status management, and employer validation workflows to ensure platform credibility and trust.	
FE-03	Job Posting & Recruitment Management	Enables HRs to create, edit, publish, close, and manage job postings and applications in a centralized recruitment workflow.	
FE-04	Candidate Profile & CV Management	Allows candidates to create profiles, upload and manage CVs, edit personal information, and maintain skills and project portfolios for recruitment purposes.	
FE-05	Job Search	Provides job searching and filtering by keyword, category, salary, experience level, location, and other recruitment criteria.	
FE-06	AI JD–CV Matching Engine	Compares candidate CVs with job descriptions and generates compatibility scores, missing skill	

		analysis, and improvement suggestions for better job matching.	
FE-07	AI Bulk Candidate Screening	Enables HRs to screen and rank large numbers of CVs automatically using AI and vector-based similarity analysis to reduce manual recruitment effort.	
FE-08	Real-Time Notification System	Sends real-time notifications for application updates, HR profile views, interview invitations, and system announcements to improve communication efficiency.	
FE-09	Feedback, Report, Violation & Moderation Management	Allows users to feedback, report spam, fake accounts, and policy violations while enabling Admins to review and moderate platform activities.	
FE-10	Recruitment Chat & Communication	Supports direct communication between HRs and candidates regarding job applications, interviews, and recruitment discussions through integrated chat features.	

4.2 Limitations & Exclusions

ID	Excluded Item	Reason
LI-01	Early-stage recruitment only (job posting, CV submission, CV screening, candidate ranking, pre-interview communication)	The system is designed to focus on the initial recruitment workflow to keep scope manageable for the project and ensure core functionality quality.
LI-02	AI-assisted evaluation and job recommendation are support tools only	Final hiring decisions must remain with HR staff and companies to ensure human responsibility and decision accuracy.
LI-03	Dependency on external AI services (OpenAI API / Gemini API)	Some AI features rely on third-party services, which may introduce cost and service limitations.
LI-04	Mobile application support	The system is developed as a web-based platform only to simplify development and focus on core features.
LI-05	Legal contracts, employment agreements, and dispute handling	The system only supports recruitment interaction and does not participate in legal or contractual matters.
LI-06	Full automation replacing recruiters	The system is designed to assist recruiters, not replace them, ensuring human oversight in hiring decisions.

5. Expected Contributions & Next Steps

5.1 Business & Operational Impact

The implementation of the Intelligent Recruitment Support System is expected to significantly reduce the manual workload of HR staff by streamlining early-stage recruitment activities such as CV collection, screening, and candidate ranking. This improvement leads to faster processing of large applicant volumes, especially for positions that attract large numbers of applicants and require extensive screening efforts. As a result, recruitment cycles are shortened, allowing companies to respond more quickly to hiring demands and reduce the risk of losing qualified candidates to competitors.

Operational efficiency is also improved through centralized management of recruitment data, reducing reliance on spreadsheets, email tracking, and fragmented tools. This helps minimize data duplication, reduce human errors, and improve overall recruitment transparency. In the long term, organizations can optimize HR resource allocation without needing additional recruitment staff for initial screening tasks.

5.2 User Experience Impact

Recruiters benefit from a more structured and less time-consuming workflow when handling candidate applications, as relevant CVs are prioritized and easier to evaluate. This reduces cognitive overload and allows HR personnel to focus more on decision-making rather than manual filtering.

Candidates experience a more relevant and responsive job application process, where AI-assisted matching helps identify positions that better align with their qualifications, skills, and experience. This increases the likelihood of receiving appropriate job opportunities and improves overall satisfaction with the recruitment process due to clearer communication and faster feedback.

5.3 Planned Next Steps After Project Completion

Following the completion and validation of the current system, future development may focus on expanding the recruitment workflow beyond the early-stage process by incorporating interview management and assessment modules. This would help create a more complete end-to-end recruitment platform.

Future versions may also explore enhanced AI capabilities, including deeper skill analysis, improved candidate-job matching accuracy, and more personalized recommendation systems. Additionally, system scalability improvements and integration with external recruitment or HR platforms may be considered to support broader enterprise adoption and real-world deployment scenarios.

[1] https://indeedinc.my.site.com/employerSupport1/s/article/360026204071?language=en_US&utm

[2] Test Validation and Construction Unit. (n.d.). *Summary of the Uniform Guidelines on Employee Selection Procedures*. University of Baltimore.

Link: https://home.ubalt.edu/tmitch/645/articles/summary_of%20uniform_guidelines-1.pdf

[3] Risavy, S. D., Robie, C., Fisher, P. A., & Rasheed, S. (2022). *Resumes vs. Application Forms: Why the Stubborn Reliance on Resumes?* *Frontiers in Psychology*, 13, Article 884205.

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